UCF Physics: AST 5765/4762: (Advanced) Astronomical Data Analysis
Practicum 7: Photometry
10/26/2025

1. (**20 points**) Write a function called `apphot()` that does aperture photometry as we described it in class. The function should take as input:
1) your object image,
2) the mask (for the good/bad pixels) you want to use,
3) the radius of your photometry aperture,
4) the inner and outer sky annulus radii, and
5) a tuple with the center coordinates of your object.

The function should follow these steps:

(a) cut out a sub-image around the approximate center of a star. You should ensure that the sub-image is big enough to also contain the sky annulus centered on the real center of the star;

(b) use function `disk()` twice to make a mask for the sky annulus;

(c) calculate the average sky pixel in the sky annulus;

(d) subtract this value from each pixel in the sub-image;

(e) use `disk()` to make a mask for the photometry aperture; and

(f) calculate the total flux in this aperture.

The function should then return a tuple containing the stellar flux and the average sky.

Bonus 3 points: you can also return in the tuple the number of bad pixels in the aperture.

2. (**10 points**) Test `apphot()`:

- Make a noiseless array containing a 2D Gaussian of σ = 2 pixels at the center of a 21×21 image.
- Run `apphot()` on it with a 6-pixel aperture and a 7–10 pixel sky annulus. -
- Print the result.
- What should it be?